

B. E. ELECTRICAL ENGINEERING 2ND YEAR 2ND SEMESTER EXAMINATION, 2025**Subject: SEQUENTIAL SYSTEMS & MICROPROCESSOR Time: 3 Hours Full Marks: 100**Use a separate Answer-script for each PartPart I (50 marks)Question
No.Question 1 is compulsory

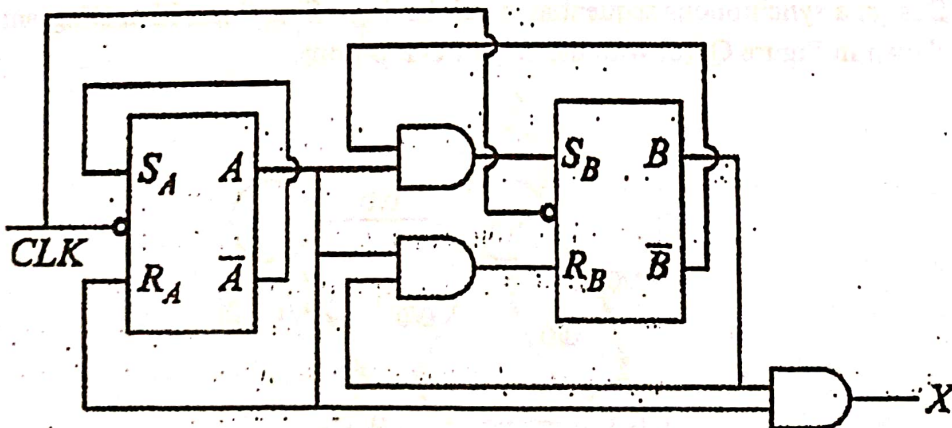
Marks

Answer Any Two questions from the rest (2×20)

Q1 Answer any Two of the following:

- (a) Explain how D Flip-Flop can be used in toggle mode. 5
- (b) What is Excitation Table for a Flip-Flop? Obtain Excitation table for J-K Flip-Flop. 5
- (c) Starting from the excitation table develop the state diagram for S-R flip-flop. 5
- (d) A $J\bar{K}$ flip-flop is constructed by using an inverter to the 'K' input. If both the inputs of the $J\bar{K}$ flip-flop are tied together to form a single input, it becomes an 'X' flip-flop. What is 'X'? Justify your conclusion. 5

- Q2 (a) What is State Synthesis Table for a Flip-flop? 2
With the help of State Synthesis Table show how SR-flip-flop can be converted to JK-flip-flop. 6
- (b) Consider, the sequential circuit shown in Figure Q2(b). It has only input CLK in the form of a fixed frequency binary pulses that triggers both the flip-flops. An output X is generated from flip-flop outputs as shown. 8
- (i) Show, with the help of state table, how the flip-flop values (or states) and output X change with input CLK . 8
- (ii) Draw the corresponding state transition diagram. 4

Figure Q2(b)

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Ref. No.: Ex/EE/PC/B/T/225/2025

- Q3 (a) (i) Show how a 4-bit Shift-Right Register can be realized using D flip-flops.
(ii) Explain, with the help of timing diagram, how a number "0100" can be entered serially into the register.
- (b) (i) What are Synchronous and Asynchronous Counters?
(ii) With the help of proper characteristic table show that the circuit shown in Figure Q3(b) combines both Synchronous and Asynchronous counter.

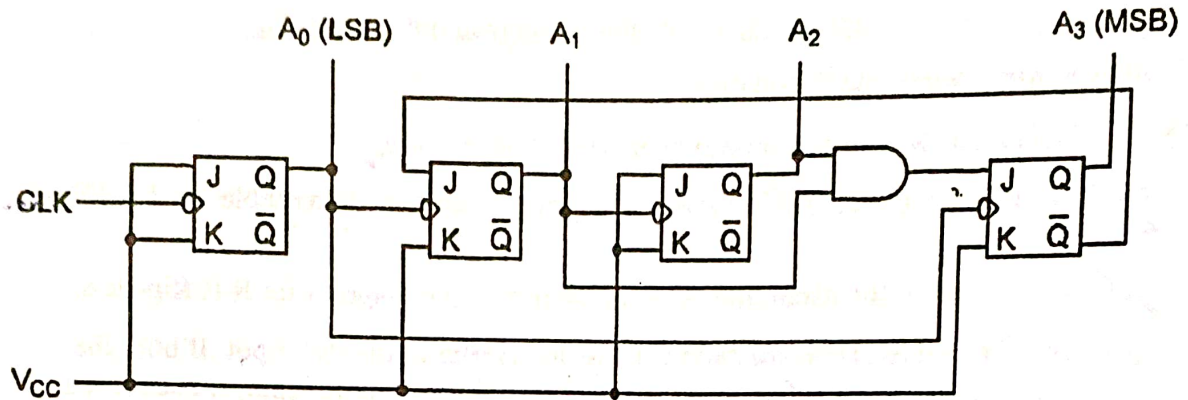


Figure Q3(b)

- Q4 (a) Enumerate, with the help of schematic diagrams, different types of Shift Register configurations, in terms of input and output interfacing, that are commercially available.
- (b) Design a modulo-4 irregular counter with the counting sequence as shown in Figure Q4(b) using D flip-flop.

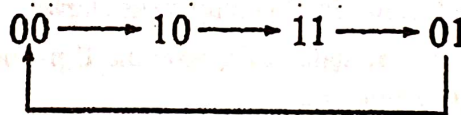


Figure Q4(b)

- (c) Design a synchronous sequential circuit having a State Transition Diagram as shown in Figure Q4(c) with the help of D flip-flops.

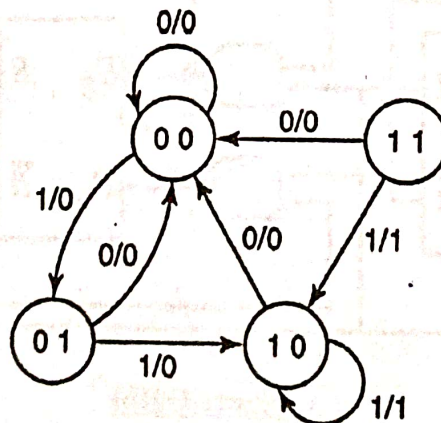


Figure Q4(c)

- Q5 (a) What is a State Transition diagram? 2
 How does the State Transition Diagram for a Moore Machine differ from that for a Mealy Machine? 2
 (b) Show how the State Transition Diagram for a Mealy Model as shown in Figures Q5(b) (i) and (ii) can be converted to their equivalent Moore Model diagrams.

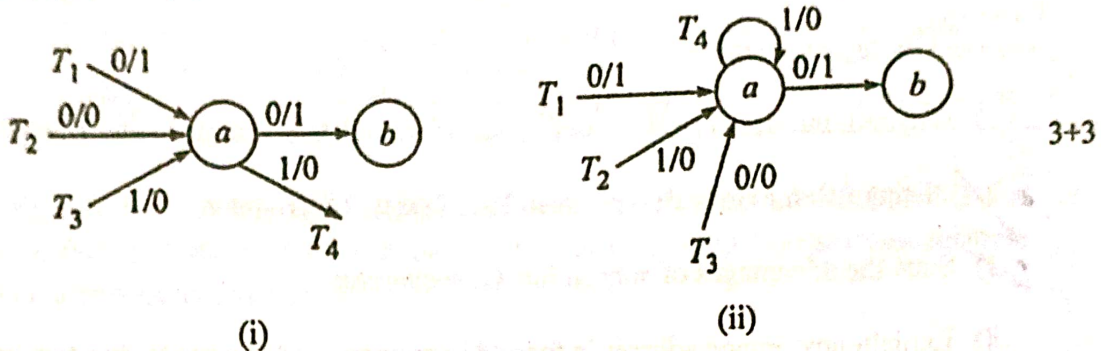


Figure Q5(b)

- (c) A sequential circuit has one input X and one output Z. The circuit is to recognize the occurrence of the sequence of bits "1101" on X by making Z equal to 1 when the previous three inputs to the circuit were "110" and current input is a 1. Otherwise, Z equals 0. 10
 Assume that Reset is applied to the direct reset inputs on its flip-flops to initialize the state of the circuit to all zeros.
 Describe, with the help of state transition table and state diagram, the synthesis process of the circuit using D flip-flop.

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B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER EXAM 2025**SEQUENTIAL SYSTEM AND MICROPROCESSORS**

Time: Three hours

(50 marks for each Part)

Full Marks: 100

Use a separate Answer-Script for each Part

PART- II**Group-A**

Answer any four questions:

4X5=20

- 1) Why unidirectional latch is used when address/data bus signals are multiplexed?
- 2) Distinguish the operation between HOLD and HALT states.
- 3) State the advantages of subroutine. Give example.
- 4) Explain how image address is formed in memory address space due to partial memory address decoding method. Give example
- 5) Distinguish between the maskable and non maskable interrupts in 8085.
- 6) Write the content of accumulator and the flag register after execution of the instructions:
 i) ADI B4H ii) CPI 11H iii) ANI 10H
 Consider the initial content of Accumulator of 8085 to be 9CH and all the bits of the flag register are reset before execution of each of the instruction.
- 7) Discuss the priority of different interrupts in 8085. How interrupts can be masked?

Group-B

Answer any three questions:

3X5=15

- 1) Illustrate a scheme for generating one wait state using wait state generator circuit and READY signal to 8085. Draw the necessary waveform also.
- 2) Show the timing diagram of RST 1 instruction in 8085 showing all necessary signal waveforms.
- 3) What are vectored interrupts? How is the address of the Interrupt Service routine calculated in vectored interrupts? Explain with an example.
- 4) The 8085 CPU is operated with clock frequency 3.77 MHz. Find out the time delay in sec can be produced by following programme:

Label	Instructions	Comments	T-States
	LXI D, 1FFFH	Load 16-bit count	10 T-states
L1:	DCX D	Decrement count	6 T-states
	MOV A, D		4 T-states
	ORA E	logically OR D and E	4 T-states
	JNZ L1	If the result $\neq$ 0, continue	10/7 T-states

- 5) Draw diagram for interfacing two RAM devices of size 2K and I/O with 8085 CPU at address 2000H and 4000H for RAM and 6000H, 6001H for I/O. Use absolute memory and I/O address decoding method.
- 6) Discuss the functions of SIM and RIM instructions in 8085.

Group-C

Answer any three questions:

3X5=15

- 1) Write an assembly language programme to delete repeated numbers from a given arrays of N elements. Store the resultant numbers in new array.
- 2) Write an assembly language programme to convert a 14-bit Hexadecimal data into 4-digit Binary coded decimal value.
- 3) Write 8085 Assembly language program to find the Fibonacci number of an 8-bit number. Draw the flow chart of the algorithm used.
- 4) Write an assembly language program in 8085 microprocessor to compute the factorial of a number (between 1 to 6). Use subroutine method.
- 5) Given an array of N nos. data already stored in memory, write an 8085 assembly language programme to find out the smallest and largest data, and store these in two consecutive memory locations.
- 6) Write an assembly language programme to generate a pulse of duration 1 ms via I/O port with address 01H. The pulse must be initiated by pressing a push button connected to RST 7.5 vector interrupt pin of 8085 CPU.
- 7) Write an 8085 assembly language programme to find the HCF of two 8-bit numbers. The numbers are stored at location 8000H and 8001H. Display the result in 8050H.